MODULE-4

COMPLEX INTEGRATION

* Line integrals in the complex plane
* Basic properties
* First evaluation method
* Second evaluation method
* Contour integrals
* Cauchy's integral theorem
* Cauchy's integral formula for derivatives of an analytic function.
* Taylor's series
* Maclaurin's series

COMPLEX INTEGRATION

Line Integrals in the Complex Plane

Complex definite integrals are called line integrals and they can be written in the form $\int_C f(z)\,dz$ where C is the given curve which is represented parametrically as $z(t) = x(t) + iy(t)$ $(a \le t \le b)$ is called the path of integration and t is increasing in the positive sense.

PROPERTIES

1) $$\int_C \left[k_1 f(z) + k_2 f(z)\right] dz = k_1 \int_C f(z)\,dz + k_2 \int_C f(z)\,dz$$

2) $$\int_{z_0}^{z_1} f(z)\,dz = -\int_{z_1}^{z_0} f(z)\,dz$$

3) $$\int_C f(z)\,dz = \int_{C_1} f(z)\,dz + \int_{C_2} f(z)\,dz$$

where C_1 and C_2 are the partitioning of C.

Indefinite integration of Analytic functions. (FIRST EVALUATION METHOD)

$$\int_{z_0}^{z_1} f(z)\,dz = F(z_1) - F(z_0)$$

Ex:- 1) $$\int_1^i \frac{dz}{z} = \left(\log z\right)_1^i = \log i - \log 1$$
$$= \log i$$

2) $$\int_0^{1+i} z^2\,dz = \left[\frac{z^3}{3}\right]_0^{1+i} = \frac{(1+i)^3}{3} = \frac{1}{3}\left[1+i^3+3i^2+3i\right]$$
$$= \frac{1}{3}\left[-2+2i\right]$$

3) $$\int_{-\pi i}^{\pi i} \cos z\,dz = \left[\sin z\right]_{-\pi i}^{\pi i} = \sin(\pi i) - \sin(-\pi i)$$
$$= \sin\pi i + \sin\pi i = 2\sin\pi i$$

4) $$\int_{8+\pi i}^{8-\pi i} e^{z/2}\,dz = \left[\frac{e^{z/2}}{1/2}\right]_{8+\pi i}^{8-\pi i} = 2\left[e^{z/2}\right]_{8+\pi i}^{8-\pi i}$$
$$= 2\left[e^{\frac{8-\pi i}{2}} - e^{\frac{8+\pi i}{2}}\right]$$

SECOND EVALUATION METHOD

(integration using the path)

Q) Find the path of integration of the following curves.

1) $z(t) = 3+i+(1-i)t \quad (0 \le t \le 3)$ (3)

Ans) $z(t) = 3+i+(1-i)t = 3+i+t-ti$

$= 3+t+(1-t)i$

We have $z(t) = x(t)+iy(t)$

$x(t) = 3+t, \quad y(t) = 1-t$

$x = 3+t, \quad t = x-3$ — (1)

$y = 1-t, \quad t = 1-y$ — (2)

Equating (1) and (2), $x-3 = 1-y$

$\Rightarrow x+y = 4$

When $t=0, x=3$ (Substitute in (1))

When $t=3, x=6$ "

When $x=3, y=1$

When $x=6, y=-2$

The path of integration is the part of straight line segment from $(3,1)$ to $(6,-2)$.

b) $z(t) = t+4t^2 i \quad (0 \le t \le 1)$

Ans) $z(t) = x(t)+iy(t) = t+4t^2 i \quad (0 \le t \le 1)$

$x(t) = t, \quad y(t) = 4t^2$

$\Rightarrow y = 4x^2$

For $t=0$ $x=0, y=0$

$t=1, x=1, y=4$

$\therefore$ The curve is the part of the parabola $y = 4x^2$ from $(0,0)$ to $(1,4)$.

3) $z(t) = 2 - 2i + \sqrt{5}\, e^{-it}, \quad 0 \le t \le 2\pi$

$= 2 - 2i + \sqrt{5}(\cos t - i\sin t)$

$= 2 + \sqrt{5}\cos t + i(-2 - \sqrt{5}\sin t)$

$x(t) = 2 + \sqrt{5}\cos t, \quad y(t) = -2 - \sqrt{5}\sin t$

$x = 2 + \sqrt{5}\cos t \Rightarrow \cos t = \dfrac{x-2}{\sqrt{5}}$

$y = -2 - \sqrt{5}\sin t \Rightarrow \sin t = -\dfrac{(y+2)}{\sqrt{5}}$

We have $\cos^2 t = \dfrac{(x-2)^2}{5}, \quad \sin^2 t = \dfrac{(y+2)^2}{5}$

$\sin^2 t + \cos^2 t = 1$

$\Rightarrow \dfrac{(x-2)^2}{5} + \dfrac{(y+2)^2}{5} = 1$

$\cancel{\Rightarrow 5 = x^2 - 4x + 4 + y^2 + 4y + 4}$

$\cancel{\Rightarrow x^2 + y^2 +} \Rightarrow (x-2)^2 + (y+2)^2 = 5$

which is a circle with centre at $(2, -2)$ and radius $\sqrt{5}$.

4) $z(t) = t + i(1-t)^3, \quad (-2 \le t \le 2)$

Ans) Let $z(t) = x(t) + iy(t)$

$$= t + i(1-t)^3 \quad (-2 \le t \le 2)$$

$x(t) = t, \quad y(t) = (1-t)^3$

$$\Rightarrow y = (1-x)^3$$

The curve is $y = (1-x)^3$ for $-2 \le t \le 2$.

5) $z(t) = \left(1 + \frac{1}{4}i\right)t, \quad 1 \le t \le 6$

Ans) Let $z(t) = x(t) + iy(t)$

$$= t + i\frac{1}{4}t, \quad 1 \le t \le 6$$

$x(t) = t, \quad y(t) = \frac{1}{4}t$

$y = \frac{x}{4} \Rightarrow x = 4y$

For $t = 1, x = 1, y = .25$

For $t = 1, x = 6 \; y = 1.5$

The path is a straight line segment $x = 4y$ from $(1, .25)$ to $(6, 1.5)$.

Q) Evaluate $\int_C e^z dz$ where C is the shortest path from $\frac{\pi}{2}i$ to πi.

Ans) $\int_C e^z dz = \int_{\pi/2 i}^{\pi i} e^z dz = \left[e^z\right]_{\frac{\pi}{2}i}^{\pi i} =$

$= e^{\pi i} - e^{\frac{\pi}{2} i} = -1-i$

a) Evaluate $\int_C \sec^2 z \, dz$ where C is any path from $\frac{\pi}{4}$ to $\frac{\pi i}{4}$

Ans) $\int_{\pi/4}^{\frac{\pi i}{4}} \sec^2 z \, dz = \left[\tan z\right]_{\frac{\pi}{4}}^{\frac{\pi i}{4}} = \tan\frac{\pi i}{4} - \tan\frac{\pi}{4}$

$= i\tan h\frac{\pi}{4} - 1$

a) Evaluate $\int_C \mathrm{Re}\, z \, dz$ from 0 to $1+2i$ along a) line segment joining $(0,0)$ and $(1,2)$

b) x-axis from 0 to 1 and vertically to $1+2i$.

Ans) Given $f(z) = \mathrm{Re}\, z$

Parametric equation of any line segment joining two points a and b is

$z(t) = (1-t)a + tb \quad (0 \le t \le 1)$

a) Let $a = (0,0)$, $b = (1,2)$

$z(t) = (1-t)(0,0) + t(1,2) = (0,0) + (t, 2t)$

$= (t, 2t)$

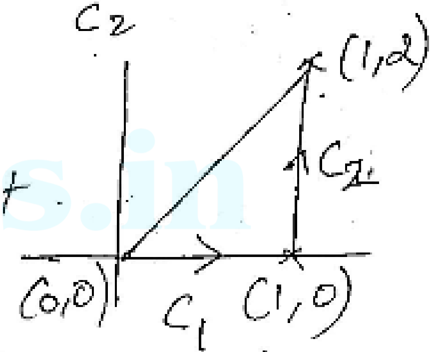
We have $\operatorname{Re} z = t$ ($\operatorname{Re} z$ is the x-co-ordinate of $z(t)$ and $\operatorname{Im} z$ is the y-co-ordinate of $z(t)$)

$z = t + 2ti$

$\operatorname{Re} z = t$, $dz = (1+2i)dt$

$$\cancel{\int_C \operatorname{Re} z \, dz = (1+2i)\int_0^1 t\,dt = (1+2i)\left(\frac{t^2}{2}\right)_0^1 = \frac{1}{2}(1+2i)}$$

b) Here $\int_C f(z)dz = \int_{C_1} f(z)dz + \int_{C_2} f(z)dz$



Along C_1

Let C_1 be the line segment joining $(0,0)$ and $(1,0)$.

The parametric representation of line segment joining $(0,0)$ and $(1,0)$ is

$$z(t) = (1-t)(0,0) + t(1,0)$$
$$= (0,0) + (t,0) = (t,0)$$

$\operatorname{Re} z(t) = t \quad (0 \le t \le 1)$

$$\therefore \int_{C_1} \operatorname{Re} z \, dz = \int_0^1 t\,dt = \left[\frac{t^2}{2}\right]_0^1 = \frac{1}{2}$$

Along C_2

$\cancel{z(t)}$ Let C_2 be the line segment joining $(1,0)$ and $(1,2)$

$$z(t) = (1-t)(1,0) + t(1,2)$$
$$= (1-t,0) + (t,2t) = (1,2t)$$

$\mathrm{Re}\, z(t) = 1$, $\quad z(t) = 1 + i2t$

$$dz = 2i\,dt \quad (0 \le t \le 1)$$

$$\int_{C_2} \mathrm{Re}\, z\, dz = \int_0^1 1 \times 2i\, dt = 2i\,[t]_0^1 = \underline{\underline{2i}}$$

$$\int_C f(z)\,dz = \int_{C_1} f(z)\,dz + \int_{C_2} f(z)\,dz$$
$$= \frac{1}{2} + 2i \;//$$

2) Evaluate $\int_C \mathrm{Re}\, z^2\, dz$ from 0 to $2+4i$ along

a) the line segment joining the points $(0,0)$ and $(2,4)$

b) x-axis from 0 to 2 and then vertically to $2+4i$.

c) the parabola $y = x^2$.

Ans) a) Equation of line segment joining the points (0,0) and (2,4) is

$$z(t) = (1-t)(0,0) + t(2,4) = (2t, 4t)$$

$$z = 2t + i4t$$

$$dz = 2dt + i4dt = (2+4i)dt$$

$$z^2 = (2t+4ti)^2 = 4t^2 + 16ti - 16t^2$$

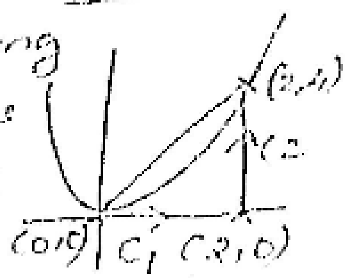
$$Re\, z^2 = 4t^2 - 16t^2 = \underline{-12t^2} \quad (0 \le t \le 1)$$

$$\int_C Re\, z^2 dz = \int_0^1 -12t^2 (2+4i) dt$$

$$= -12(2+4i)\left[\frac{t^3}{3}\right]_0^1 = \frac{-12(2+4i)}{3}$$

$$= -4(2+4i) = \underline{-8-16i}$$

b) Let C_1 be the line segment joining the points 0 to 2 and C_2 be the line segment joining the points (2,0) to (2,4)



$$\therefore \int_C f(z)dz = \int_{C_1} f(z)dz + \int_{C_2} f(z)dz$$

Along C_1

Equation of line segment joining $(0,0)$ and $(2,0)$ is

$$z(t) = (1-t)(0,0) + t(2,0) = (2t, 0)$$

$$z(t) = 2t, \quad dz = 2dt$$

$$z^2 = 4t^2, \quad 0 \le t \le 1$$

$$\therefore \int_{C_1} \operatorname{Re} z^2\, dz = \int_0^1 4t^2 \times 2dt = 8\int_0^1 t^2 dt$$

$$= 8\left[\frac{t^3}{3}\right]_0^1 = \frac{8}{3}$$

<u>Along C_2</u>

Equation of line segment joining $(2,0)$ and $(2,4)$ is given by

$$z(t) = (1-t)(2,0) + t(2,4)$$

$$= (2(1-t), 0) + (2t, 4t)$$

$$= (2-2t, 0) + (2t, 4t)$$

$$= (2, 4t)$$

$$z = 2 + i4t, \quad dz = 4i\,dt$$

$$z^2 = (2+4it)^2 = 4 + 16it - 16t^2$$

$$\operatorname{Re} z^2 = 4 - 16t^2$$

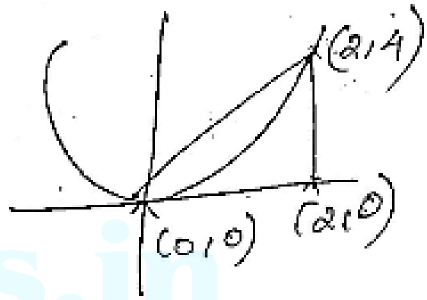
$$\int_{C_2} \operatorname{Re} z^2\, dz = 4i\int_0^1 (4-16t^2)\,dt$$

$$= 4i\left[4t - \frac{16t^3}{3}\right]_0^1 = 4i\left[4 - \frac{16}{3}\right]$$

$$= 4i \times -\frac{4}{3} = \frac{-16i}{3}$$

$$\therefore \int_C \operatorname{Re} z^2\, dz = \frac{8}{3} - \frac{16}{3}i$$

c) Along parabola $y = x^2$



Let $x = t$, $y = t^2$

Parametric representation of parabola is

$$z = z(t) = t + it^2, \quad 0 \le t \le 2.$$

$$dz = 1 + 2it\,dt = (1 + 2it)\,dt$$

$$z^2 = (t + it^2)^2 = t^2 + 2it^3 - t^4$$

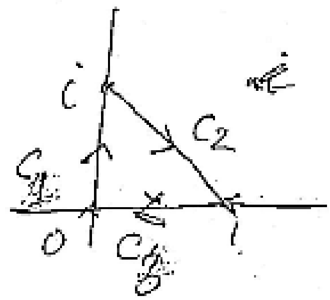
$$\operatorname{Re} z^2 = t^2 - t^4$$

$$\int_C \operatorname{Re} z^2\, dz = \int_0^2 (t^2 - t^4)(1 + 2it)\,dt$$

$$= \left[\frac{t^3}{3} - \frac{t^5}{5} + 2i\left(\frac{t^4}{4} - \frac{t^6}{6}\right)\right]_0^2 = -\frac{56}{15} - \frac{40}{3}i$$

Q) Evaluate $\int_C \text{Im}\, z^2 dz$ where C is the path joining the triangle with vertices $0, 1, i$ in clockwise direction.

Ans) $\int_C f(z)dz = \int_{C_1} f(z)dz + \int_{C_2} f(z)dz + \int_{C_3} f(z)dz$



~~Ans~~ **Along C1.**

Equation of line segment joining 0 and i is $z(t) = (1-t)(0,0) + t(0,1)$

$= (0,0) + (0,t) = (0,t)$

$z(t) = ti \quad , dz = i\,dt \quad (z(t))^2 = -t^2$

$\text{Im}\, z^2 = 0 \quad 0 \le t \le 1$

$\int_{C_1} \text{Im}\, z^2 dz = \int_0^1 \cancel{0\,dt = 0}$

Along C2

Equation of line segment joining $(0,1)$ and $(1,0)$ is

$z(t) = (1-t)(0,1) + t(1,0) = (0,1-t) + (t,0)$

$= (t, 1-t)$

$z(t) = t + i(1-t)$

$\cancel{z'(t) = (1-i)dt}$ $dz = (1-i)dt$

$\operatorname{Im} z^2 dz = \operatorname{Im}(t+i(1-t))^2 = 2t(1-t)$

$\int_C \operatorname{Im} z^2 dz = \int_0^t 2t(1-t)(1-i)dt$

$= 2(1-i)\left[\frac{t^2}{2} - \frac{t^3}{3}\right]_0^1 = \frac{1-i}{3}$

Along C_3

Equation of line segment joining $(1,0)$ and $(0,0)$ is

$z(t) = (1-t)(1,0) + t(0,0) = (1-t, 0)$

$z(t) = (1-t) + 0i$, $\operatorname{Im}(z(t))^2 = 0$.

$\therefore \int_{C_3} \operatorname{Im}(z(t))^2 dt = \int_{C_3} \operatorname{Im} z^2 dz = 0$

$\therefore \int_C f(z)dz = \int_{C_1} f(z)dz + \int_{C_2} f(z)dz + \int_{C_3} f(z)dz$

$= 0 + \frac{1-i}{3} + 0 = \frac{1-i}{3}$

CAUCHY'S INTEGRAL THEOREM

Contour integral

A closed path that does not intersect itself is called a simple closed path or contour and an integral over such a path is called a contour integral.

CAUCHY'S THEOREM

If $f(z)$ is analytic in a simply connected domain D, then for every contour C in D

$$\oint_C f(z)\,dz = 0.$$

Ex:- 1) $\int e^z dz = 0$, since e^z is analytic function for all z.

2) ~~2)~~ $\oint_C \sin z\, dz = 0$, since $\sin z$ is analytic for all z.

3) ~~3)~~ $\oint_C \sec z\, dz$ over the unit circle $C: |z| = 1$

Let C be a unit circle, the function $\sec z = \frac{1}{\cos z}$ is not analytic at $z = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$ and all these points lie outside C, So $\oint_C \sec z \, dz = 0$.

4) 4) Consider the integral $\oint \frac{dz}{z^2+4}$

$\frac{1}{z^2+4}$ is not analytic at $z = \pm 2i$

which lie outside C.

So $\oint_C \frac{dz}{z^2+4} = 0$.

5) 6) Evaluate $\int_C \tan z \, dz$, $C: |z|=1$

Ans) Here $f(z) = \tan z = \frac{\sin z}{\cos z}$

$f(z)$ is analytic except at points where

$\cos z = 0 \Rightarrow z = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$

All these points lie outside the circle $|z|=1$.

$\therefore$ By Cauchy's theorem, $\int_C \tan z \, dz = 0$.

6) 7) Evaluate $\int_C \frac{e^z}{z^2+9} dz$, $C: |z| = 2$.

Ans) Here $f(z) = e^z$, $z^2+9=0 \Rightarrow z_0 = \pm 3i$

Both the points lie outside the circle

$|z|=2$.

$\therefore$ By Cauchy's theorem, $\int_C \frac{e^z}{z^2+9} dz = 0$.

7) Evaluate $\int_C e^{\sin z^2} dz$, $C : |z| = 1$

The integrand $e^{\sin z^2}$ is analytic at all points inside and on the circle $|z| = 1$. Therefore, by Cauchy's theorem

$$\int_C e^{\sin z^2} dz = 0.$$

8) Evaluate $\int_C z^2 + z + 4 \, dz$, $C : |z - i| = 1$

The integrand $f(z) = z^2 + z + 4$ is analytic at all points inside and on the circle $|z - i| = 1$. By Cauchy's theorem,

$$\int_C (z^2 + z + 4) \, dz = 0.$$

9) Evaluate $\int_C \frac{z^2 + 5}{z - 3} dz$ where C is $|z - i| = 1$

Ans) Here $f(z) = z^2 + 5$ and $z_0 = 3$, since $f(z)$ is analytic at all points inside and on the circle $|z = 1|$ and $z_0 = 3$ lie outside C,

By Cauchy's theorem, $\int_C \frac{z^2 + 5}{z - 3} dz = 0.$

10) Evaluate $\int_C \frac{3z^2+7z+1}{z+1}\, dz$ where $C: |z+i|=1$

Ans) Here $f(z) = 3z^2+7z+1$ and $z_0 = -1$. Since $f(z)$ is analytic at all points inside and on the circle $|z+i|=1$ and $z_0=-1$ lie outside C. By Cauchy's theorem,

$$\int_C \frac{3z^2+7z+1}{z+1}\, dz = 0.$$

11) Evaluate $\int_C \frac{dz}{4z-1}$ around the unit circle $|z|=1$.

Ans) Here $f(z) = \frac{1}{4z-1}$ and is not analytic at $z = 1/4$ and it lies inside $|z|=1$

$$\int_C \frac{dz}{4z-1} = \int_C \frac{1}{4(z-\frac{1}{4})}\, dz = \frac{1}{4} \times 2\pi i = \frac{\pi i}{2}$$

<u>Result:-</u>

$$\oint_C \frac{1}{z-z_0}\, dz = 2\pi i$$

12) Evaluate $\oint \frac{dz}{z-2i}$ where C is the circle $|z|=\pi$ counter clockwise.

Ans) The point $z=2i$ lie outside $C: |z|=\pi$

Hence $\oint \frac{dz}{z-2i} = 0.$

CAUCHY'S INTEGRAL THEOREM FOR MULTIPLY CONNECTED DOMAINS

Let $f(z)$ be analytic in a domain D bounded by non intersecting simple closed curves $C_1, C_2, \ldots, C_n$ where $C_1, C_2 \ldots C_n$ lie inside C.

$$\oint_C f(z)\,dz = \oint_{C_1} f(z)\,dz + \oint_{C_2} f(z)\,dz + \cdots + \oint_{C_n} f(z)\,dz$$

CAUCHY'S INTEGRAL FORMULA

Let $f(z)$ be analytic in a simply connected domain D. Then for any point z_0 in D and any simple closed path that encloses z_0 in D

$$\int_C \frac{f(z)}{z - z_0} = 2\pi i f(z_0)$$

Examples

1) Evaluate $\int \frac{\cos 2z}{4z}$ around the unit circle

Ans) $f(z) = \cos 2z$ and $z_0 = 0$. Since $f(z)$ is analytic at all points within and on the circle $|z - i| = 1$ and $z_0 = 0$ lies inside $|z| = 1$

$\therefore$ By Cauchy's integral formula

$$\int \frac{\cos 2z}{4z} = \frac{1}{4} \times 2\pi i\, f(z_0)$$

$$= \frac{1}{4} \times 2\pi i \times \cos 0 = \frac{\pi i}{2}$$

2) $\int_C \frac{z^2}{4z-i}$ around the unit circle $|z|=1$.

Ans) Here $f(z) = z^2$ and $z_0 = \frac{i}{4}$. Since $f(z)$ is analytic at all points except at $z_0 = \frac{i}{4}$ lies inside the circle.

So $$\int \frac{z^2}{4\left(z-\frac{i}{4}\right)} = \frac{1}{4} \times 2\pi i\, f\left(\frac{i}{4}\right)$$

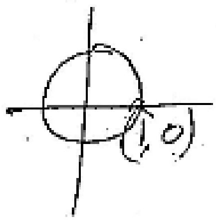
$$= \frac{1}{4} \times 2\pi i \times \left(\frac{i}{4}\right)^2 = \frac{-2\pi i}{64} = \frac{-\pi i}{32}$$

3) Evaluate $\int_C \frac{z^2-z+1}{z-1}\,dz$ where C is the circle (i) $|z|=1$ (ii) $|z| = \frac{1}{2}$

Ans) i) Here $f(z) = z^2 - z + 1$ and $z_0 = 1$. Since $f(z)$ is analytic at all points inside and on the circle $|z|=1$, z_0 lies on the circle

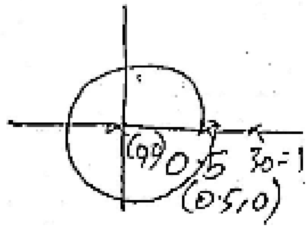
So $$\int \frac{z^2-z+1}{z-1}\,dz = 2\pi i\, f(1) = 2\pi i(1-1+1)$$

$$= 2\pi i$$



(ii) $f(z) = z^2 - z + 1$ and $z_0 = 1$. Since $f(z)$ is analytic at all points inside and on the circle $|z| = \frac{1}{2}$, $z_0 = 1$ lies outside $|z| = \frac{1}{2}$

Hence $\int_C \frac{z^2 - z + 1}{z - 1} \, dz = 0.$



4) Evaluate $\int_C \frac{\cos \pi z}{z - 1} \, dz$ ~~one~~ where C is the circle $|z| = 3$.

Ans) Here $f(z) = \cos \pi z$ and $z_0 = 1$. Since $f(z)$ is analytic at all points in and on the circle.

Here $z_0 = 1$ lies inside the circle $|z| = 3$.

$$\int_C \frac{\cos \pi z}{z - 1} \, dz = 2\pi i f(z_0) = 2\pi i f(1)$$

$$= 2\pi i \times \cos \pi$$

$$= -2\pi i$$

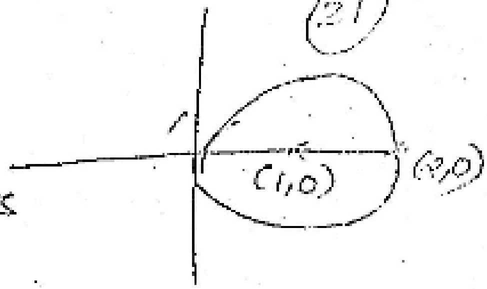
5) Evaluate $\int_C \frac{3z^2 + z}{z^2 - 1}$ where C is the circle $|z - 1| = 1$.

Ans) $f(z) = 3z^2 + z$ and $z_0 = 1$ and ~~radius~~

$|z - 1| = 1$ is a circle with centre at $(1, 0)$

and radius 1.

The function $f(z) = 3z^2 + z$ is analytic at all points inside and on the circle.



$z_0 = 1$ lies inside the circle.

So $\int_C \frac{3z^2+z}{z^2-1}\,dz = \int \frac{3z^2+z/z+1}{z-1}\,dz$

$\boxed{\text{Here we take } f(z) = \frac{3z^2+z}{z+1}}$

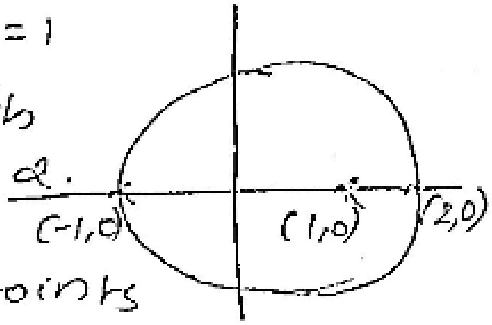
$$= 2\pi i f(1) = 2\pi i \left[\frac{3z^2+z}{z+1}\right]_{z=1} = 2\pi i\left(\frac{4}{2}\right)$$

$$= \underline{4\pi i}$$

6) $\int_C \frac{z+2}{z-1}\,dz$ where $C: |z-1| = 2$

Ans) Here $f(z) = z+2$ and $z_0 = 1$

$C: |z-1| = 2$ is a circle with centre at $(1,0)$ and radius 2.



$f(z)$ is analytic at all points inside and on the circle.

$z_0 = 1$ is a point which lies inside $|z-1| = 2$

$$\int_C \frac{z+2}{z-1}\,dz = 2\pi i f(1) = 2\pi i (z+2)$$

$$= 2\pi i \times 3 = 6\pi i$$

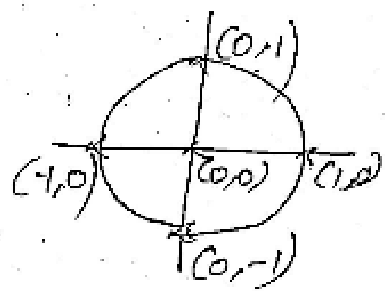
7) Evaluate $\int_C \frac{e^{2z}}{z-i}$ where C is the unit circle $|z|=1$.

Ans) $f(z)=e^{2z}$ and $z_0=i$.

$f(z)$ is analytic at all points inside and on the circle.

$z_0=i$ is the point $(0,1)$ which lies ~~outside~~ on C.

$$\therefore \int_C \frac{e^{2z}}{z-i}\,dz = 2\pi i f(i)$$
$$= 2\pi i e^{2i}$$



8) Evaluate $\int_C \frac{\sin\pi z^2+\cos\pi z^2}{(z-1)(z-2)}$ where C is the circle $|z|=3$.

Ans) Here $f(z)=\sin\pi z^2+\cos\pi z^2$

Since $f(z)$ is analytic at all points inside and on the circle $|z|=3$, which is a circle with centre at $(0,0)$ and radius 3.

The points $z=1$ and $z=2$ both lie inside the circle $|z|=3$.

So splitting into partial fractions,

$$\frac{1}{(z-1)(z-2)} = \frac{A}{z-1} + \frac{B}{z-2}$$

$$\frac{1}{(z-1)(z-2)} = \frac{A(z-2)+B(z-1)}{(z-1)(z-2)}$$

We get $A=-1$ when $z=1$

$B=1$ when $z=2$.

$$\frac{1}{(z-1)(z-2)} = \frac{-1}{z-1} + \frac{1}{z-2}$$

$$\int_C \frac{\sin\pi z^2+\cos\pi z^2}{(z-1)(z-2)}\,dz = -\int \frac{\sin\pi z^2+\cos\pi z^2}{z-1}\,dz + \int \frac{\sin\pi z^2+\cos\pi z^2}{z-2}\,dz$$

$$= -2\pi i f(1) + 2\pi i f(2)$$

$$= -2\pi i(0+(-1)) + 2\pi i(0+1)$$

$$= 2\pi i + 2\pi i = \underline{4\pi i}$$

9) Use Cauchy's integral formula to evaluate $\oint_C \frac{z^2+1}{z^2-1}\,dz$ counter clockwise around (i) $|z-1|=1$ (ii) $|z+1|=1$.

Ans) $f(z) = \frac{z^2+1}{(z+1)(z-1)}$

(i) $z_0 = 1$, take $f(z) = \frac{z^2+1}{z+1}$

$$\int_C \frac{z^2+1}{z^2-1}\,dz = \int_C \frac{\frac{z^2+1}{z+1}}{(z-1)}\,dz = 2\pi i f(1)$$

$$= 2\pi i$$

(ii) $z_0 = -1$, take $f(z) = \frac{z^2+1}{z-1}$

$$\int_C \frac{z^2+1}{z^2-1}\,dz = \int_C \frac{\frac{z^2+1}{z-1}}{(z+1)}\,dz = 2\pi i f(-1)$$

$$= -2\pi i$$

10) Evaluate $\oint_C \frac{\log(z+1)}{z^2+1}$ where C is $|z-i|=1.4$

Ans) The non analytic points are $z^2+1=0 \Rightarrow$ $z=i$ lies inside C and $z=-i$ lies outside C

Choose $f(z) = \frac{\log(z+1)}{z+i}$ and $z_0 = i$

$$\oint_C \frac{\log(z+1)}{z^2+1}\,dz = 2\pi i f(i) = 2\pi i \frac{\log(1+i)}{2i}$$

$$= \pi \log(1+i) = \pi \log(\sqrt{2}+i\pi/4)$$

CAUCHY'S INTEGRAL FORMULA FOR HIGHER ORDER DERIVATIVES OF AN ANALYTIC FUNCTION

If $f(z)$ is analytic with derivatives of all orders in a domain D of a complex plane and C is any simple closed path in D that encloses z_0 and oriented in counter clockwise direction,

then $$\int_C \frac{f(z)}{(z-z_0)^{n+1}}\,dz = \frac{2\pi i}{n!} f^{(n)}(z_0)$$

where $f^{n}(z_0)$ is the nth derivative of $f(z)$ at z_0.

Examples

1) Evaluate $\int_C \frac{\sinh 2z}{\left(z-\frac{1}{2}\right)^4}\,dz$ around the unit circle $|z|=1$ counter clockwise.

Ans) The function $f(z)=\sinh 2z$ is analytic at all points inside and on the circle $|z|=1$ and the point $z_0=1/2$ lies inside $|z|=1$

By Cauchy's integral formula,

$$\int \frac{\sinh 2z}{(z-1/2)^4} = \frac{2\pi i}{3!}\left[\frac{d^3}{dz^3}\sinh 2z\right]_{z=1/2}$$

$$= \frac{2\pi i}{6}\left[8\cosh 2z\right]_{z=\frac{1}{2}} = \frac{8\pi i}{3}\cosh 1$$

2) Evaluate $\int_C \frac{e^z \cos z}{\left(z-\frac{\pi}{4}\right)^3}\, dz$ counter clockwise around the unit circle.

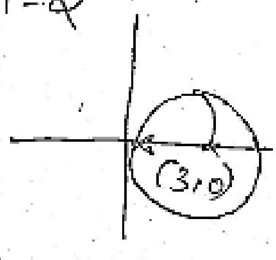
Ans) Here $f(z)=e^z\cos z$ is analytic at all points inside and on the circle $|z|=1$ and $z_0=\frac{\pi}{4}$ lies inside $|z|=1$. By Cauchy's integral formula,

$$\int_C \frac{e^z\cos z}{\left(z-\frac{\pi}{4}\right)^3}\, dz = \frac{2\pi i}{2!}\left[\frac{d^2}{dz^2} e^z\cos z\right]_{z=\frac{\pi}{2}}$$

$$= \pi i\left[-2e^z\cos z\right]_{z=\frac{\pi}{2}} = \sqrt{2}\,\pi i e^{\pi/4}$$

3) Evaluate $\oint_C \frac{\log z}{(z-4)^2}\, dz$ counter clockwise around the circle $|z-3|=2$.

Ans) Here $f(z)=\log z$ is analytic at all points inside and on the circle $|z-3|=2$ and $z_0=4$ lies inside the circle.



$\therefore$ By Cauchy's integral formula,

$$\int_C \frac{\log z}{(z-4)^2}\,dz = \frac{2\pi i}{1!} f'(4)$$

$$= 2\pi i\left(\frac{1}{z}\right) = 2\pi i\left(\frac{1}{4}\right) = \frac{\pi i}{2}$$

4) Evaluate $\int_C \frac{e^{2z}}{(z+1)^4}\,dz$ where $C: |z| = 2$.

Ans) Here $f(z) = e^{2z}$ and $z_0 = -1$. Since $f(z)$ is analytic at all points inside and on the circle $|z| = 2$ and $z_0 = -1$ lies inside $|z| = 2$.

$$\therefore \int_C \frac{e^{2z}}{(z+1)^4}\,dz = \frac{2\pi i}{3!} f^3(e^{2z})$$

$$= \frac{2\pi i}{6} \times 8e^{2(-1)} = \frac{8\pi i}{3} e^{-2}$$

5) Evaluate $\int_C \frac{e^{3z}}{(4z-\pi i)^3}$ around the unit circle.

Ans) Here $f(z) = e^{3z}$ and $z_0 = \frac{\pi i}{4}$

Here $f(z)$ is analytic at all points in and on the circle.

$C: |z| = 1$ is a circle with centre $(0,0)$ and radius 1.

Here $z_0 = \frac{\pi i}{4}$ ie $(0, .785)$ lies inside

So $\int \frac{e^{3z}}{(4z-\pi i)^3} = \frac{1}{4^3} \oint_C \frac{e^{3z}}{\left(z-\frac{\pi i}{4}\right)^3} dz$

$= \frac{1}{64} \times \frac{2\pi i}{2!} \left[\frac{d^2}{dz^2} e^{3z}\right]_{z=\frac{\pi}{4}} = \frac{-9\pi(1+i)}{64\sqrt{2}}$

6) Evaluate $\int_C \frac{e^{2z}}{(z+1)^4} dz$ where C is $|z|=2$

Ans) Here $f(z) = e^{2z}$, $z_0 = -1$. Since $f(z)$ is analytic within and on the circle $|z|=2$ and $z_0=-1$ lies inside C. By Cauchy's integral formula

$$\int_C \frac{e^{2z}}{(z+1)^4} dz = \frac{2\pi i}{3!} \left[\frac{d^3}{dz^3} e^{2z}\right]_{z=-1}$$

$$= \frac{2\pi i}{6} \left[8e^{2z}\right]_{z=-1} = \frac{8\pi i}{3e^2}$$

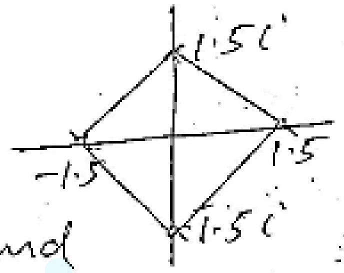
7) Evaluate $\oint_C \frac{\tan \pi z}{z^2} dz$ where C is the ellipse $16x^2+y^2=1$ counter clockwise.

Ans) Here $f(z) = \tan \pi z$ and $z_0=0$ lies inside the ellipse. By Cauchy's integral formula,

$$\oint_C \frac{\tan \pi z}{z^2} dz = 2\pi i \left[\frac{d}{dz} \tan \pi z\right]_{z=0}$$

$$= 2\pi i\, \pi \left[\sec^2 \pi z\right]_{z=0} = \underline{\underline{2\pi^2 i}}$$

8) Evaluate $\oint_C \frac{\log(z+3)}{(z-2)(z+1)^2} dz$ where C is the boundary of the square with vertices $\pm 1.5, \pm 1.5i$ counter clockwise.



Ans) The point $z=-1$ lies inside C and $z=2$ lies outside C.

So we choose $f(z) = \frac{\log(z+3)}{z-2}$ and

$z_0 = -1$.

$$\oint_C \frac{\log(z+3)/z-2}{(z+1)^2} = 2\pi i \left[\frac{d}{dz} \frac{\log(z+3)}{z-2}\right]_{z=-1}$$

$$= 2\pi i \left[(z-2)\frac{1}{z+3} + \log(z+3)\right]_{z=-1} = 2\pi i \left[\frac{-3}{2} + \log 2\right]$$

9) Evaluate $\int_C \frac{z^3-2z+1}{(z-i)^2}$ where C is $|z|=2$

Ans) Here $f(z) = z^2 - 2z + 1$ is analytic at all points except at $z=i$, which lies inside C.

Ans) The integrand is analytic at all points except at $z=i$ which lie inside C.

$$\int_C \frac{z^3-2z+1}{(z-i)^2} = 2\pi i\left[\frac{d}{dz}(z^3-2z+1)\right]_{z=i}$$

$$=2\pi i\left[3z^2-2\right]_{z=i} = 2\pi i\left[3i^2-2\right] = 2\pi i(-5)$$

$$= -10\pi i$$

10) Evaluate $\int_C \frac{e^z}{(z-1)^2(z^2+4)}\,dz$ where C is the circle $|z-1|=2$.

Ans) The non analytic points are $z=1$ and $z=\pm 2i$. In the contour $|z-1|=2$, $z=1$ lies inside and $z=\pm 2i$ lies outside.

Ans) $f(z)=\frac{e^z}{z^2+4}$ and $z_0=1$

$$=2\pi i\left[\frac{d}{dz}\frac{e^z}{z^2+4}\right]_{z=1} = 2\pi i\left[\frac{(z^2+4)e^z-e^z 2z}{(z^2+4)^2}\right]$$

$$=2\pi i\left(\frac{3e}{25}\right) = \frac{6e\pi i}{25}$$

POWER SERIES

The series representation in powers of $(z-z_0)$ of the form,

$$\sum_{n=0}^{\infty} a_n (z-z_0)^n = a_0 + a_1(z-z_0) + a_2(z-z_0)^2 + \dots$$

where z is a complex variable is called a power series. The coefficients $a_0, a_1, \dots$ are the coefficients of the series and z_0 is a complex constant is called the centre of the series.

TAYLOR'S SERIES

Let $f(z)$ be analytic in a domain D and let $z = z_0$ be any point in D. Then there exists precisely one Taylor series

$$f(z) = f(z_0) + \frac{(z-z_0)}{1!} f'(z_0) + \frac{(z-z_0)^2}{2!} f''(z_0) + \dots + \frac{(z-z_0)^n}{n!} f^n(z_0) + R_n(z)$$

with centre z_0 that represents $f(z)$.

The Taylor's series with centre $z_0 = 0$ is called Maclaurin series.

Maclaurin's series

The Maclaurin's series expansion of $f(z)$ is given by

$$f(z) = f(z_0) + \frac{z}{1!} f'(z_0) + \frac{z^2}{2!} f''(z_0) + \cdots$$

The radius of convergence of the series is $R = \lim_{n \to \infty} \left| \frac{a_n}{a_{n+1}} \right|$

If $\left| \frac{a_n}{a_{n+1}} \right| \to \infty$ then $R = 0$.

A power series with non zero radius of convergence is the Taylor series of its sum.

Find the Maclaurin series of the following functions.

a) $f(z) = e^z$

Ans) $f(z) = e^z$, $f(0) = 1$

$f'(z) = e^z$, $f'(0) = 1$

$f''(z) = e^z$, $f''(0) = 1$

Maclaurin's series of $f(z)$ is

$$f(z) = f(0) + \frac{z}{1!} f'(0) + \frac{z^2}{2!} f''(0) + \cdots$$

$$e^z = 1 + z + \frac{z^2}{2!} + \cdots = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

b) $f(z) = \cos z, \quad f(0) = 1$

$f'(z) = -\sin z, \quad f'(0) = 0$

$f''(z) = -\cos z, \quad f''(0) = -1$

$f'''(z) = \sin z, \quad f'''(0) = 0$

$f^{iv}(z) = \cos z, \quad f^{iv}(0) = 1$

$$\therefore \cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} + \cdots$$

c) $f(z) = \sin z \quad f(0) = 0.$

$f'(z) = \cos z \quad f'(0) = 1$

$f''(z) = -\sin z \quad f''(0) = 0.$

$f'''(z) = -\cos z \quad f'''(0) = -1$

$f^{iv}(z) = \sin z, \quad f^{iv}(0) = 0.$

$f^{v}(z) = \cos z, \quad f^{v}(0) = -1$

$$\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} + \cdots$$

d) $f(z) = \cosh z$

Ans) $f(z) = \cosh z$, $f(0) = 1$

$f'(z) = \sinh z$, $f'(0) = 0$

$f''(z) = \cosh z$, $f''(0) = 1$

$f'''(z) = \sinh z$, $f'''(0) = 0$.

$f^{iv}(z) = \cosh z$, $f^{iv}(0) = 1$

$\therefore \cosh z = 1 + \frac{z^2}{2!} + \frac{z^4}{4!} + \cdots$

e) $f(z) = \sinh z$

$f(z) = \sinh z$ $\quad f(0) = 0$.

$f'(z) = \cosh z$ $\quad f'(0) = 1$

$f''(z) = \sinh z$ $\quad f''(0) = 0$.

$f'''(z) = \cosh z$ $\quad f'''(0) = 1$

$f^{iv}(z) = \sinh z$ $\quad f^{iv}(0) = 0$

$\therefore f(z) = \sinh z = z + \frac{z^3}{3!} + \frac{z^5}{5!} + \cdots$

f) $f(z) = \log(1+z)$

$f(z) = \log(1+z)$, $\quad f(0) = \log 1 = 0$.

$f'(z) = \frac{1}{1+z}$, $\quad f'(0) = 1$.

$f''(z) = \frac{-1}{(1+z)^2}$, $f''(0) = -1$

$f'''(z) = \frac{2}{(1+z)^3}$, $f'''(0) = 2$.

$\therefore \log(1+z) = z - \frac{z^2}{2!} + \frac{2z^3}{3!} - \cdots$

$= z - \frac{z^2}{2} + \frac{z^3}{3} - \cdots$

g) $-\log f(z) = -\log(1-z)$

Ans) $\log(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \cdots$

Replacing ~~z~~ z by $-z$ and multiplying both sides by -1, we have

$-\log(1-z) = z + \frac{z^2}{2} + \frac{z^3}{3} + \cdots$

$\log \frac{1}{1-z} = z + \frac{z^2}{2} + \frac{z^3}{3} + \cdots \quad (|z|<1)$

h) $\log\left(\frac{1+z}{1-z}\right)$ (By above problem)

Ans) $\log(1+z) + \log\frac{1}{1-z} = 2z + \frac{2z^3}{3} + \frac{2z^5}{5} + \cdots$

$\log\left(\frac{1+z}{1-z}\right) = 2\left(z + \frac{z^3}{3} + \frac{z^5}{5} + \cdots\right)$

Q) Find the Maclaurins series of arc tan z

Ans) $f(z) = \arctan z$, $f(0)=0$

$f'(z) = \frac{1}{1+z^2}$, $f'(0)=1$

$f''(z) = -2z+4z^3-6z^5+\dots$, $f''(0)=0$

$f'''(z) = -2+12z^2-30z^4+\dots$ $f'''(0)=-2$

$f^{iv}(z) = 24z-120z^3$, $f^{iv}(0)=0.$

$f^{v}(z) = 24-360z^2+\dots$, $f^{v}(0)=24$

$$\arctan z = z - \frac{2z^3}{3!} + \frac{24z^5}{5!} + \dots$$

$$= z - \frac{z^3}{3} + \frac{z^5}{5} + \dots$$

Q) Expand the function $f(z)=1/z$ in Taylor series with centre $z_0=i$. Find the radius of convergence.

Ans) $$\frac{1}{z} = \frac{1}{z-i+i} = \frac{1}{i}\left[1+\frac{z-i}{i}\right]^{-1}$$

$$= \frac{1}{i}\left[1-\frac{z-i}{i}+\left(\frac{z-i}{i}\right)^2-\left(\frac{z-i}{i}\right)^3+\dots\right]$$

which converges when $\left|\frac{z-i}{i}\right|<1$ or

$|z-i|=i$

Q) Find the Taylor series expansion of $f(z)=\frac{1}{1+z}$ with centre $z_0=i$.

Ans) $$\frac{1}{1+z}=\frac{1}{1+z+i-i}=\frac{1}{1-i\left(1+\frac{z+i}{1-i}\right)}=\frac{1}{1-i}\left[1+\frac{z+i}{1-i}\right]^{-1}$$

$$=\frac{1}{1-i}\left[1-\frac{z+i}{1-i}+\left(\frac{z+i}{1-i}\right)^2+\cdots\right]$$

$$=\frac{1}{1-i}-\frac{(z+i)}{(1-i)^2}+\frac{(z+i)^3}{(1-i)^3}+\cdots$$

$$=\frac{1+i}{2}-\frac{i}{2}(z+i)+\frac{1-i}{4}(z+i)^2+\cdots$$

Q) Find the Taylor series expansion of $f(z)=\cos z$ about $z=\pi$. Find the radius of convergence.

Ans) $$f(z)=\cos z=\cos(z-\pi+\pi)=-\cos(z-\pi)$$

$$=-\left[1-\frac{(z-\pi)^2}{2!}+\frac{(z-\pi)^4}{4!}+\cdots\right]$$

$$=-1+\frac{(z-\pi)^2}{2!}-\frac{(z-\pi)^4}{4!}-\cdots$$

$$R=\lim_{n\to\infty}\left|\frac{a_n}{a_{n+1}}\right|=\lim_{n\to\infty}\frac{(2n+1)!}{(2n)!}=\infty.$$

Q) Find the Taylor series expansion of $f(z) = \log z$ with 3 as center.

Ans) $f(z) = \log z$, $f(3) = \log 3$

$f'(z) = \frac{1}{z}$, $f'(3) = \frac{1}{3}$

$f''(z) = \frac{-1}{z^2}$, $f''(3) = \frac{-1}{9}$

$f'''(z) = \frac{2}{z^3}$ $f'''(3) = \frac{2}{27}$

$$\log z = \log 3 + \frac{z-3}{3} - \frac{(z-3)^2}{2 \cdot 3^2} + \frac{(z-3)^2}{3 \cdot 3^2} - \dots$$

Q) Find the Taylor series expansion of $f(z) = \cosh(z - \pi i)$ about $z_0 = \pi i$:

Ans) $\cosh(z-\pi i) = \frac{e^{z-\pi i} + e^{-(z-\pi i)}}{2}$

$$= \frac{1}{2}\left[\left[1 + \frac{z-\pi i}{1!} + \frac{(z-\pi i)^2}{2!} + \frac{(z-\pi i)^3}{3!} + \dots\right] + \left[1 - \frac{z-\pi i}{1!} + \frac{(z-\pi i)^2}{2!} - \frac{(z-\pi i)^3}{3!} + \dots\right]\right]$$

$$= \frac{1}{2}\left[2 + 2\frac{(z-\pi i)^2}{2!} + 2\frac{(z-\pi i)^4}{4!} + \dots\right]$$

$$= 1 + \frac{(z-\pi i)^2}{2!} + \frac{(z-\pi i)^4}{4!} + \dots$$

QUESTION BANK

1) Find the path of integration for the following curves.

a) $z(t) = 5e^{-it}$ $\left(0 \le t \le \frac{\pi}{2}\right)$

b) $z(t) = 2\cos t + i\sin t$ $(0 \le t \le 2\pi)$

c) $z(t) = t + i(1-t)^2$ $(-1 \le t \le 1)$

2) Evaluate $\int_C \mathrm{Re}\, z\, dz$ where C is the shortest path from $1+i$ to $5+5i$.

3) Evaluate $\int_C \mathrm{Re}\, z^2\, dz$ clockwise around the boundary of the square with vertices $0, i, 1+i, 1$.

4) Evaluate $\int_C \mathrm{Re}\, z\, dz$ from 0 to $2+8i$ along $y = 2x^2$.

5) Integrate the following function around the unit circle $|z| = 1$.

a) $\dfrac{e^{2z}}{\pi z - i}$ b) $\dfrac{z \sin z}{2z - 1}$

6) Integrate $\dfrac{z^2}{z^2 - 1}$ by Cauchy's integral formula counter clockwise around the following circles.

a) $|z+1| = \frac{3}{2}$ b) $|z-1-i| = \frac{\pi}{2}$ c) $|z+5-5i| = 7$

7) Evaluate $\int_C \frac{e^z}{z-2}\,dz$ counter clockwise around the circle $|z|=3$.

8) Evaluate $\int_C \frac{z^3-6}{2z-i}\,dz$ counter clockwise around the circle $|z|=1$.

9) Evaluate $\oint_C \frac{e^{3z}}{3z-i}\,dz$ counter clockwise around $|z|=1$.

10) Evaluate $\oint_C \frac{e^z}{z-2i}\,dz$ counter clockwise around $|z-2i|=4$.

11) Evaluate $\oint_C \frac{\cos z}{2z}\,dz$ where C is $|z|=1/2$.

12) Evaluate $\oint_C \frac{1}{z^2-1}\,dz$ where C is

a) $|z+1| = 1/2$ b) $|z-1| = \pi/2$

13) Integrate counter clockwise around the unit circle.

a) $\oint \frac{\sin 2z}{z^4}\,dz$ b) $\oint_C \frac{z^6}{(2z-1)^6}\,dz$ c) $\oint_C \frac{e^z \cos z}{\left(z-\frac{\pi}{4}\right)^3}\,dz$

d) $\oint_C \frac{\sinh 2z}{(z-1/2)^4}\,dz$ e) $\oint_C \frac{dz}{(z-2i)^2\left(z-\frac{i}{2}\right)^2}$

14) Evaluate $\oint_C \frac{e^{z^2}}{z(z-2i)^2}\,dz$ where C is $|z-3i|=2$

~~15) Evaluate~~ $\oint_C \frac{4z^3-6}{\cancel{z(z+1-i)^2}}\,dz$

15) Evaluate $\oint_C \frac{\tan \pi z}{(z-1)^2}\,dz$ clockwise around $|z-1|=0.1$

16) Evaluate $\oint_C \frac{\log z}{(z-2i)^2}\,dz$ counter clockwise around $|z-1|=1/2$.

17) Find the Taylor series expansion of $f(z)=\frac{1}{z^2}$ about the point $z=2$.

18) Find the Taylor series expansion with centre z_0 for the following functions.

a) $\sin z$, $z_0=\frac{\pi}{4}$ b) $\cos z$, $z_0=\frac{\pi}{2}$

c) $\frac{z-1}{z+1}$, $z_0=0$ d) $\frac{z}{z+2}$, $z_0=1$.